Introduction

3 minutes

Artificial Intelligence (AI) refers to systems designed to perform tasks that typically require human intelligence—such as reasoning, problem-solving, perception, and language understanding.

An **AI application** is a software solution that uses AI techniques—such as computer vision, speech, and information extraction—to perform tasks that typically require human-like intelligence. These applications can understand, reason, learn, and respond to inputs in a way that feels more adaptive than traditional software.

AI applications are powered by *machine learning* (ML) models, which are mathematical systems trained to recognize patterns in data and make predictions or generate outputs. ML models are the engines inside an AI application. When you interact with an AI application, the model performs *inference*, meaning it applies what it learned during training to new input.

AI applications are:

- *Model-powered*: They use trained models to process inputs and generate outputs, such as text, images, or decisions.
- *Dynamic*: Unlike static programs, AI apps can improve over time through retraining or fine-tuning.

Some examples of AI applications for different industries include:

- **Healthcare**: AI-powered diagnostic tools that analyze medical images (such as X-rays or MRIs) and help doctors detect diseases more accurately and quickly.

- **Finance**: Fraud detection systems that use AI to monitor transactions in real time and identify suspicious activity, helping prevent financial crimes.
- **Retail**: Personalized recommendation engines that analyze customer behavior and preferences to suggest products, improving the shopping experience.
- **Manufacturing**: Predictive maintenance solutions that use AI to monitor equipment and forecast when machines are likely to fail, reducing downtime and maintenance costs.
- **Education**: Intelligent tutoring systems that adapt to each student's learning style and pace, providing customized feedback and support to enhance learning outcomes.

In this module, you learn how Microsoft enables you to build AI applications with the latest technology, securely, and at scale. While the model is the engine, AI applications also need security, networking, hosting, data storage, application logic, and user interfaces. Microsoft provides all the infrastructure and services needed to support enterprise-scale AI development. The module gives you a foundation in how Azure streamlines AI application development, integrates with Microsoft Foundry, and enables rapid innovation.

> **Note**
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> We recognize that different people like to learn in different ways. You can choose to complete this module in video-based format or you can read the content as text and images. The text contains greater detail than the videos, so in some cases you might want to refer to it as supplemental material to the video presentation.

Next unit: Understand Azure

Next